

ID2216 Project Proposal

Canvas Group 5 Members

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App Proposal

Name: PubTrail

Short Description

PubTrail is a collaborative mobile application that allows users to record and share their pub crawl experiences.

Users can create a pub crawl session and add the different places they visit during the night by selecting locations on a map. At each stop, they can upload photos and notes to document their experience.

Each pub crawl generates a QR code that friends can scan to join the same crawl session and contribute their own photos and locations. This allows groups to collaboratively document the same pub crawl in real time.

Users can also explore a map view that displays public pub crawls created by others, allowing them to discover popular routes and nightlife spots.

To add a playful element, the app includes a small gyroscope-based mini-game where users attempt to keep their phone balanced as a humorous “sobriety test”.

Frameworks that will be used

The application will be developed as a mobile app using React Native:

React Native (Expo)

We will use React Native with Expo to build the cross-platform mobile application. Expo simplifies development by providing built-in support for many device features such as the camera, sensors, and media library, which are needed for our app.

Expo also allows rapid testing on real devices during development.

TypeScript

Using TypeScript will make the application easier to maintain and reduce runtime errors, especially as the application grows and multiple developers collaborate on the codebase.

Redux

We will use Redux for state management in the application. Redux will manage shared state across the app, such as:

- the current pub crawl session
- list of visited locations
- photos uploaded during a crawl
- user participation in a crawl

Using Redux ensures that multiple screens can easily access and update the same data without complicated prop passing.

APIs Used

OpenStreetMap API

Used to display maps and select pub locations. This will allow users to:

- browse a map
- search for locations
- add stops based on real-world coordinates.

Firebase API

Firebase will be used as the backend for storing data and media. Services used:

- Firebase Firestore for storing pub crawl data
- Firebase Cloud Storage for uploading and retrieving photos

Device APIs (via Expo)

React Native (Expo) allows access to mobile device features:

- Camera API, used for taking photos at each pub stop & scanning QR codes for joining pub crawls
- Media Library API, used for uploading photos from the phone
- Device Motion / Gyroscope API, used for the sobriety mini-game

App-Specific Data (stored in Firestore)

Pub Crawl

- crawl ID
- crawl name
- creator
- list of participants
- list of stops

Stop

- pub name
- coordinates
- timestamp
- notes
- uploaded photos

Photo

- image file (points to Cloud Firebase)
- uploader user

User

- user id/username/email
- avatar photo

Screen Mockups

